P9

Practical 9: Prediction Using Linear Regression

Aim

To perform **prediction using Linear Regression in R programming**.

Theory

Linear Regression is a statistical method used to **find the relationship between two variables**.
The relationship is represented by the equation:

$$y = ax + b$$

Where:

- **y** → Response variable (dependent variable)
- **x** → Predictor variable (independent variable)
- **a** → Slope of the line
- **b** → Intercept

If the relationship is **linear**, the graph forms a **straight line**.
If the exponent of variables is not 1, it forms a **curve (non-linear relationship)**.
Example:
Predict **weight** of a person based on **height**.

Steps Involved

1. Collect sample data of **height and weight**.
2. Create a **linear regression model** using the **lm() function in R**.
3. Find the **coefficients of the regression equation**.
4. View the **summary of the model** to check prediction accuracy.
5. Predict the **weight of a new person** using the **predict() function**.

Algorithm / Procedure

1. Define predictor and response variables.
2. Apply **lm() function** to create regression model.
3. Display the regression coefficients.
4. Print summary of the model.
5. Predict output for new input data.
6. Plot regression graph.

```
# Input data: height and weight
x <- c(151, 174, 138, 186, 128, 136, 179, 163, 152, 131)
y <- c(63, 81, 56, 91, 47, 57, 76, 72, 62, 48)

# Create linear regression model
relation <- lm(y ~ x)

# Print the model coefficients
print(relation)

# Display summary of the relationship
print(summary(relation))
```

```
# Predict weight for a person with height 170
a <- data.frame(x = 170)
result <- predict(relation, a)
print(result)

# Create PNG file for graph
png(file = "linearregression.png")

# Plot regression graph
plot(y, x,
    col = "blue",
    main = "Height & Weight Regression",
    abline(lm(x ~ y)),
    cex = 1.3,
    pch = 16,
    xlab = "Weight in Kg",
    ylab = "Height in cm")

# Save the graph
dev.off()
```